

Bridging Physics

Tutorial 1

2026

1. The position of a particle is given by $x(t) = At^2 + Be^{\beta t}$, where A , B and β are constants. At time $t = 0$, the particle is at $x = 1.0$ m and moving with velocity $v = 4.0$ m/s. Its velocity is observed to be 10 m/s at $t = 0.2$ s. What is the acceleration of the particle at $t = 1.0$ s? (879.06 m/s²)

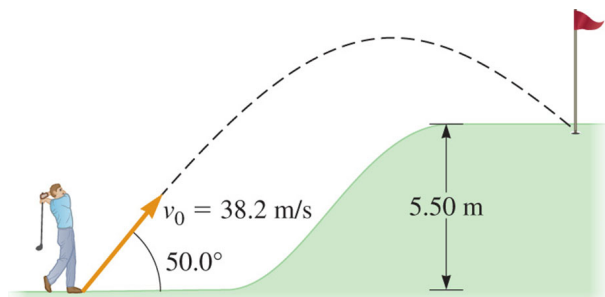
2. During your summer internship for an aerospace company, you are asked to design a small research rocket. The rocket is to be launched from rest from the earth's surface and is to reach a maximum height of 960 m above the earth's surface. The rocket's engines give the rocket an upward acceleration of 16.0 m/s² during the time T that they fire. After the engines shut off, the rocket is in free fall. Ignore air resistance. What must be the value of T in order for the rocket to reach the required altitude? (6.75 s)

3. An object is moving along the x -axis. At time $t = 0$ it has velocity $v_{0x} = 20.0$ m/s. Starting at time $t = 0$ it has acceleration $a_x = -Ct$, where C has units of m/s³.
 - (i) What is the value of C if the object stops in 8.00 s after $t = 0$? $\left(\frac{5}{8}\right)$
 - (ii) For the value of C calculated in part (i), how far does the object travel during the 8.00 seconds? (106.67 m)

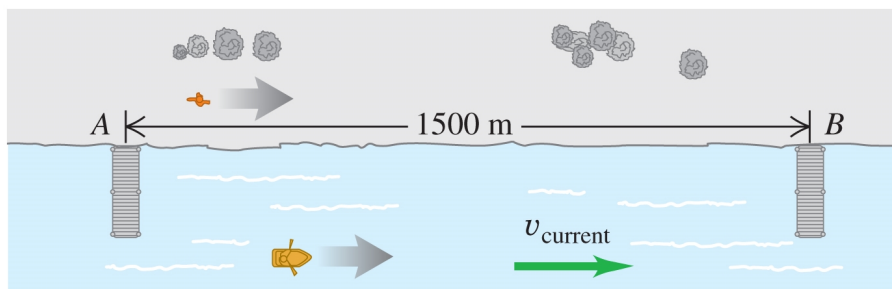
4. A boy 12.0 m above the ground in a tree throws a ball for his dog, who is standing right below the tree and starts running the instant the ball is thrown. If the boy throws the ball horizontally at 8.50 m/s,
 - (i) how fast must the dog run to catch the ball just as it reaches the ground, and
 - (ii) how far from the tree will the dog catch the ball? (8.5 m/s, 13.3 m)

Now suppose that the boy throws the ball upward at 60.0° above the horizontal, but all else is the same. Repeat parts (i) and (ii). (4.25 m/s, 10.57 m)

5. A golfer hits his shot at an angle of 50° , giving the ball an initial speed of 38.2 m/s (refer to figure). The ball lands on the elevated green 5.50 m above the initial position near the hole, and stop immediately.



- (i) How much time passed while the ball was in the air? (5.78 s)
 - (ii) How far did the ball travel horizontally before landing? (141.85 m)
 - (iii) What was the peak height reached by the ball? (43.69 m)
6. A ball swings counterclockwise in a vertical circle at the end of a rope 1.50 m long. When the ball is 36.9° past the lowest point on its way up, its (total) acceleration is $\vec{a} = (-22.5\hat{i} + 20.2\hat{j}) \text{ m/s}^2$. For that instant,
- (i) sketch a vector diagram showing the components of its acceleration,
 - (ii) determine the magnitude of its centripetal acceleration, and (29.66 m/s^2)
 - (iii) determine the speed and velocity of the ball. (6.67 m/s , $5.33\hat{i} + 4\hat{j}$)
7. Two piers, A and B , are located on the river; B is 1500 m down-stream from A (refer to figure). Two friends must make round trips from pier A to pier B and return. One of them rows a boat at a constant speed of 4.00 km/h relative to the water, while the other walks on the shore at a constant speed of 4.00 km/h . The velocity of the river is 2.8 km/h in the direction from A to B . Will the two friends take the same time for their round trip? Justify. If not, how much time does it take each person to make the round trip?



8. A bird flies in the xy -plane with a velocity vector given by $\vec{v} = (\alpha - \beta t^2)\hat{i} + \gamma t\hat{j}$, with $\alpha = 2.4 \text{ m/s}$, $\beta = 1.6 \text{ m/s}^3$, and $\gamma = 4.0 \text{ m/s}^2$. The positive y -direction is vertically upward. At time $t = 0$ the bird is at the origin.

- (i) Calculate the position and acceleration vectors of the bird as functions of time. $\left(\left(\frac{12}{5}t - \frac{8}{15}t^3 \right) \hat{i} + 2t^2 \hat{j}, -3.2t \hat{i} + 4 \hat{j} \right)$
- (ii) What is the bird's altitude (y-coordinate) as it flies over $x = 0$ for the first time after $t = 0$? (9 m)
9. You throw a ball vertically upward from the roof of a tall building. The ball leaves your hand at point O with an upward velocity of 15.0 m/s and is then in free fall. On its way down, it passes by point O and continues to descend to the bottom of the building. Find
- the ball's position and velocity 4.00 s after leaving your hand; (-18.4 m, -24.2 m/s)
 - the maximum height reached; (11.48 m)
 - the ball's acceleration when it is at its maximum height;
 - the ball's velocities when it is 5.00 m above point O; (11.27 m/s)
 - the time when the ball has fallen 5.00 m below point O. (3.36 s)
10. A batter hits a baseball so that it leaves the bat at speed $v_0 = 37.0$ m/s at an angle of $\alpha_0 = 53.1^\circ$.
- Find the position of the ball and its velocity at $t = 2.00$ s. ((44.43, 39.58) m, (22.2, 9.99) m/s)
 - Find the time when the ball reaches the highest point of its flight, and its height h at this time. (3.02 s, 44.67 m)
 - Find the horizontal range R , i.e., the horizontal distance from the starting point to where the ball hits the ground. Assume that the height of the initial hit point on the ball from the ground was relatively negligible. (134.18 m)
11. The speed of a particle moving on a circle 2.0 m in radius increases at the constant rate of 4.4 m/s². At the instant when the magnitude of the (total) acceleration is 6.0 m/s², what is the speed of the particle? (2.86 m/s)
12. A river flows due south with a speed of 2.5 m/s. You steer a motorboat across the river; your velocity relative to the water is 4.2 m/s due east. The river is 900 m wide.
- What is your velocity (magnitude and direction) relative to the Earth? (4.89 m/s, 30.76° South of East)
 - How much time is required to cross the river? (214.29 s)
 - How far south of your starting point will you reach the opposite bank? (535.72 m)

Please solve all the questions in advance.

- End -

Formula List

Logarithm:

$$\log a + \log b = \log(a \times b)$$

$$\log a - \log b = \log\left(\frac{a}{b}\right)$$

$$(\log x) > 0, \text{ if } x > 1.$$

$$(\log 1) = 0.$$

$$(\log x) < 0, \text{ if } 0 < x < 1.$$

$$(\log x) \text{ is undefined if } x < 0.$$

Trigonometry:

$$\sin = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin}{\cos}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Differentiation:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \times \frac{d}{dx}(f(x))$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(f(x) \times g(x)) = f(x) \times \left(\frac{d}{dx}g(x)\right) + \left(\frac{d}{dx}f(x)\right) \times g(x)$$

Integration:

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

Vector:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

$$\vec{A} \bullet \vec{B} = |\vec{A}| \times |\vec{B}| \times \cos \theta$$

Linear kinematic equations:

$$v_f = v_i + at$$

$$\Delta x = v_i t + \frac{1}{2} at^2$$

$$\Delta x = \frac{(v_i + v_f)}{2} \times t$$

$$v_f^2 = v_i^2 + 2a\Delta x$$

Simple harmonic motion:

$$x(t) = r_0 \cos(\theta_0 + \omega_0 t)$$

$$v_x(t) = -r_0 \omega_0 \sin(\theta_0 + \omega_0 t)$$

$$a_x(t) = -r_0 \omega_0^2 \cos(\theta_0 + \omega_0 t) \\ = -\omega_0^2 x(t)$$

Projectile motion:

$$y = -\frac{g}{2v_{0x}^2}(x - x_0)^2 + \frac{v_{0y}}{v_{0x}}(x - x_0) + y_0, \text{ or}$$

$$y = x \tan \theta - \frac{gx^2}{2v_0^2}(1 + \tan^2 \theta)$$

Uniform circular motion:

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T}$$

$$v = r\omega$$

$$a_c = r\omega^2 = \frac{v^2}{r}$$

Non-uniform circular motion:

$$a_t = \frac{dv}{dt}$$

$$a = \sqrt{a_c^2 + a_t^2}$$

Relative velocity:

$$\vec{V}_{P/A} = \vec{V}_{P/B} + \vec{V}_{B/A}$$

Newton's second law:

$$\sum \vec{F} = m\vec{a}$$

Newton's third law:

$$\vec{F}_{12} = -\vec{F}_{21}, \text{ and the two forces} \\ \text{act on the opposite body.}$$

Frictional force:

$$f_s \leq \mu_s n, \text{ and impending motion} \\ \text{occurs when } f_s = \mu_s n.$$

$$f_k = \mu_k n$$

Translational kinetic energy:

$$KE_{\text{translational}} = \frac{1}{2}mv^2$$

Gravitational potential energy:

$$\Delta PE = mg(y_2 - y_1) \\ = mgh$$

Spring

$$F_{\text{spring}} = -kx$$

$$\text{Elastic } PE_{\text{spring}} = \frac{1}{2}kx^2$$

Work-energy theorem:

$$W = \Delta KE = KE_f - KE_i$$

Work done (constant force):

$$W = \vec{F} \bullet \Delta \vec{r} = F \Delta r \cos \theta$$

Work done (varying force):

$$W = \int_{x_i}^{x_f} \vec{F} \bullet d\vec{r}$$

Power:

$$P = \frac{dW}{dt} = F \times v$$

Linear momentum:

(also valid for y and z)

$$p_x = mv_x$$

$$F_x = \frac{dp_x}{dt}$$

$$F_{\text{average}, x} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} F_x(t) dt$$

Conservation of linear momentum:

$$\vec{p}_{1,i} + \vec{p}_{2,i} = \vec{p}_{1,f} + \vec{p}_{2,f}$$

Elastic collision:

$$v_{A1} - v_{B1} = -(v_{A2} - v_{B2})$$

$$v_{A2} = \left(\frac{m_A - m_B}{m_A + m_B} \right) v_{A1} + \left(\frac{2m_B}{m_A + m_B} \right) v_{B1}$$

$$v_{B2} = \left(\frac{m_B - m_A}{m_A + m_B} \right) v_{B1} + \left(\frac{2m_A}{m_A + m_B} \right) v_{A1}$$

Center of mass

(also valid for y and z)

$$x_{\text{cm}} = \frac{1}{M} \sum mx, \text{ where } M = \sum m, \text{ or}$$

$$= \frac{1}{M} \int x dm$$

$$v_{\text{cm}, x} = \frac{1}{M} \sum mv_x$$

$$a_{\text{cm}, x} = \frac{1}{M} \sum ma_x$$

Simple Harmonic Motion:

$$a_x = -\omega^2 x$$

$$\omega^2 = \frac{k}{m}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$v_{\max} = \sqrt{\frac{k}{m}} A$$

$$a_{\max} = \frac{k}{m} A$$

Retarding force:

$$F = -bv$$

Rotation

$$\theta = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

$$v_{\text{tangential}} = r\omega$$

$$a_{\text{centripetal}} = r\omega^2 = \frac{v^2}{r}$$

$$a_{\text{tangential}} = r\alpha$$

Rotational kinematic equations:

$$\omega_f = \omega_i + \alpha t$$

$$\Delta\theta = \omega_i t + \frac{1}{2} \alpha t^2$$

$$\Delta\theta = \frac{(\omega_i + \omega_f)}{2} \times t$$

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$$

Moment of inertia:

$$I = \sum mr^2$$

$$I = \int r^2 dm$$

Parallel axis theorem

$$I = I_{cm} + Md^2$$

Rotational kinetic energy:

$$KE_{\text{rotational}} = \frac{1}{2} I \omega^2$$

Moment of inertia:

$$I = \sum m_i r_i^2$$

$$I = \int r^2 dm$$

Moment of inertia of hoop or thin cylindrical shell:

$$I_{CM} = MR^2$$

Moment of inertia of hollow cylinder:

$$I_{CM} = \frac{1}{2} M (R_1^2 + R_2^2)$$

Moment of inertia of solid cylinder:

$$I_{CM} = \frac{1}{2} MR^2$$

Moment of inertia of rectangular plate:

$$I_{CM} = \frac{1}{12} M (a^2 + b^2)$$

Moment of inertia of thin rod about its CM:

$$I_{CM} = \frac{1}{12} ML^2$$

Moment of inertia of thin rod about its end:

$$I = \frac{ML^2}{3}$$

Moment of inertia of solid sphere:

$$I_{CM} = \frac{2}{5} MR^2$$

Moment of inertia of spherical shell:

$$I_{CM} = \frac{2}{3} MR^2$$

Torque and Moment of force:

$$\tau = F \times r \times \sin \theta$$

$$\sum \tau = I\alpha$$

Work due to rotation:

$$\Delta W = \tau \times \Delta \theta$$

$$W = \int_{\theta_1}^{\theta_2} \tau \, d\theta$$

Power due to rotation:

$$P = \tau \times \omega$$

Pure rolling:

$$v_{cm} = r\omega$$

$$a_{cm} = r\alpha$$

$$KE_{\text{total}} = \frac{1}{2} Mv_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$

Conservation of Angular momentum:

$$\underbrace{\sum I\omega}_{\text{Before collision}} = \underbrace{\sum I\omega}_{\text{After collision}}$$

Avogadro's number:

$$N_A \approx 6.022 \times 10^{23} \text{ particles/mole}$$

Ideal gas constant:

$$R \approx 8.314 \text{ J mole}^{-1} \text{ K}^{-1}$$

Boltzmann constant:

$$k = \frac{R}{N_A} \approx 1.381 \times 10^{-23} \text{ J/K}$$

Ideal gas equation:

$$pV = nRT$$

Speed of a molecule

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

Mole

$$n = \frac{N}{N_A}$$

$$n = \frac{m}{M}$$

Absolute temperature

$$T = T_{\text{Celsius}} + 273.15$$

Specific heat c and molar heat C

$$Q = Mc(T_2 - T_1)$$

$$= nC(T_2 - T_1)$$

Latent heat L

$$Q = L \times M$$

First law of thermodynamics

$$\Delta U = Q - W$$

Internal energy for N molecules

$$\text{monatomic: } U = N \times \frac{3}{2} kT$$

$$\text{diatomic: } U = N \times \frac{5}{2} kT$$

Relationship between C_p and C_v for an ideal Gas

$$C_p - C_v = R$$

Isochoric process

$$\text{monatomic: } C_v = \frac{3}{2} R$$

$$\text{diatomic: } C_v = \frac{5}{2} R$$

C_p is undefined for isochoric process.

Isobaric process

$$\text{monatomic: } C_p = \frac{5}{2}R$$

$$\text{diatomic: } C_p = \frac{7}{2}R$$

C_v is undefined for isobaric process.

Isothermal process

$$W = nRT \ln \left(\frac{V_f}{V_i} \right)$$

Adiabatic process

$$\gamma = \frac{C_p}{C_v}$$

$$\text{monatomic: } \gamma = \frac{5}{3}$$

$$\text{diatomic: } \gamma = \frac{7}{5}$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$p_1^{1-\gamma} T_1^\gamma = p_2^{1-\gamma} T_2^\gamma$$

Thermal efficiency of a heat engine

$$e = 1 - \frac{|Q_C|}{|Q_H|}$$

Coefficient of performance of a heat pump for cooling

$$\text{COP} = \frac{|Q_C|}{|Q_H| - |Q_C|}$$

Entropy

$$\Delta S = \int_{\text{initial state}}^{\text{final state}} \frac{dQ}{T} \geq 0$$

Coulomb's Law:

$$\mathbf{F} = \frac{k q_1 q_2}{r^2} \text{ where}$$

$$k = 8.9875 \times 10^9 \text{ Nm}^2/\text{C}^2$$

Electric field:

$$\mathbf{E} = \frac{\mathbf{F}}{q_0}$$

Electric field for point charge:

$$\mathbf{E} = \frac{kq}{r^2}$$

Electric field at a distance along central axis of charged ring:

$$\mathbf{E} = \frac{kQx}{(x^2 + a^2)^{3/2}}$$

Electric potential:

$$V = \frac{U}{q_0}$$

Electric potential in uniform field:

$$\Delta V = -Ed$$

Electric potential for point charge:

$$V = \frac{kq}{r}$$

Electric potential at a distance along central axis of charged ring:

$$V = \frac{kQ}{\sqrt{x^2 + a^2}}$$

Electric potential energy (for multiple point charges):

$$U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

Electric current:

$$I = \frac{dQ}{dt}$$

$$I_{av} = nq v_d A$$

Ohm's Law:

$$I = \frac{\Delta V}{R}$$

Resistance of ohmic conductor:

$$R = \rho \frac{L}{A}$$

Resistivity:

$$\rho = \rho_0 [1 + \alpha(T - T_0)]$$

Electric power:

$$P = I\Delta V = I^2 R = \frac{\Delta V^2}{R}$$

Electric energy:

$$U = P\Delta t$$

Terminal voltage:

$$\Delta V = \mathcal{E} - Ir$$

Resistors in series:

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

Resistors in parallel:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Kirchhoff's Junction Rule:

$$\sum I_{in} = \sum I_{out}$$

Kirchhoff's Loop Rule:

$$\sum \Delta V = 0$$

Magnetic force on a moving charge:

$$\vec{F}_B = q\vec{v} \times \vec{B} \sin \theta$$

Magnetic force on a current carrying conductor:

$$\vec{F} = I\vec{L} \times \vec{B} \sin \theta$$

Radius of moving charged particle in uniform magnetic field:

$$r = \frac{mv}{qB}$$

Angular speed:

$$\omega = \frac{qB}{m}$$

Period:

$$T = \frac{2\pi m}{qB}$$

Velocity selector:

$$\vec{v} = \frac{\vec{E}}{\vec{B}}$$

Magnetic field due to long straight conductor:

$$B = \frac{\mu_0 I}{2\pi a} \text{ where}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

Magnetic field at center of circular wire loop:

$$B = \frac{\mu_0 I}{2R}$$

Magnetic field at a distance along central axis of current-carrying ring:

$$B = \frac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}}$$

Magnetic force between parallel conductors:

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi a}$$

Magnetic field inside toroid:

$$B = \frac{\mu_0 NI}{2\pi r}$$

Magnetic field inside solenoid:

$$B = \mu_0 nI$$

Magnetic flux:

$$\Phi_B = BA \cos \theta$$

Faraday's Law:

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

Motional emf:

$$\mathcal{E} = \frac{Blv}{R}$$

Power input:

$$P = F_{app} v = I l B v$$

Mass

Electron: 9.1094×10^{-31} kg

Proton: 1.6726×10^{-27} kg

Neutron: 1.6749×10^{-27} kg

Charge

Electron: -1.6022×10^{-19} C

Proton: $+1.6022 \times 10^{-19}$ C

Neutron: 0 C